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## Index of Species Information

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- [Introductory](#)
  - [Distribution and Occurrence](#)
  - [Management Considerations](#)
  - [Botanical and Ecological Characteristics](#)
  - [Fire Ecology](#)
  - [Fire Effects](#)
  - [References](#)
- 

### Introductory

#### SPECIES: *Pinus monticola*

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##### AUTHORSHIP AND CITATION :

Griffith, Randy Scott. 1992. *Pinus monticola*. In: Fire Effects Information System, [Online]. U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station, Fire Sciences Laboratory (Producer). Available: <https://www.fs.fed.us/database/feis/plants/tree/pinmot/all.html> [2021, April 23].

##### ABBREVIATION :

PINMOT

##### SYNONYMS :

*Pinus monticola* var. *minima* Lemmon  
*Pinus strobus* L. var. *monticola* (Dougl. ex D. Don) Nutt.  
*Strobus monticola* Rydb.

##### SCS PLANT CODE :

PIM03

##### COMMON NAMES :

western white pine  
mountain white pine  
Idaho white pine  
silver pine

##### TAXONOMY :

The currently accepted scientific name of western white pine is *Pinus monticola* Dougl. ex D. Don (Pinaceae) [[11](#),[38](#)]. There are two recognized varieties: *P. m.* var. *minima* Lemmon and *P. m.* var. *monticola* [[38](#)]. There are no subspecies or forms.

Western white pine hybridizes with Balkan pine (*P. peuce*), blue pine (*P. griffithii*), eastern white pine (*P. strobus*), southwestern white pine (*P. strobiformis*), and limber pine (*P. flexilis*) [[11](#)].

**LIFE FORM :**

Tree

**FEDERAL LEGAL STATUS :**

No special status

**OTHER STATUS :**

NO-ENTRY

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## DISTRIBUTION AND OCCURRENCE

### SPECIES: *Pinus monticola*

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**GENERAL DISTRIBUTION :**

Western white pine occurs in the Pacific Northwest. The northern boundary of its range is at Quesnel Lake, British Columbia, latitude 52 deg. 30 min. N., and the southern boundary is at Tulare County, California, latitude 35 deg. 51 min. N. The western boundary is marked by the Pacific Coast, and the eastern boundary is at Glacier National Park, Montana. Western white pine reaches its greatest size and best stand and commercial development in northern Idaho and adjacent parts of Montana, Washington, and British Columbia [[11](#)].

**ECOSYSTEMS :**

FRES22 Western white pine  
FRES23 Fir - spruce  
FRES24 Hemlock - Sitka spruce  
FRES25 Larch  
FRES26 Lodgepole pine

**STATES :**

CA ID MT NV OR WA AB BC

**BLM PHYSIOGRAPHIC REGIONS :**

1 Northern Pacific Border  
2 Cascade Mountains  
4 Sierra Mountains  
8 Northern Rocky Mountains

**KUCHLER PLANT ASSOCIATIONS :**

K001 Spruce - cedar - hemlock forest  
K002 Cedar - hemlock - Douglas-fir forest  
K003 Silver fir - Douglas-fir forest  
K004 Fir - hemlock forest  
K005 Mixed conifer forest  
K007 Red fir forest  
K012 Douglas-fir forest  
K013 Cedar - hemlock - pine forest  
K014 Grand fir - Douglas-fir forest

**SAF COVER TYPES :**

205 Mountain hemlock  
206 Engelmann spruce - subalpine fir  
207 Red fir

210 Interior Douglas-fir  
212 Western larch  
213 Grand fir  
215 Western white pine  
218 Lodgepole pine  
224 Western hemlock  
226 Coastal true fir - hemlock  
227 Western redcedar - western hemlock  
228 Western redcedar  
229 Pacific Douglas-fir  
230 Douglas-fir - western hemlock  
231 Port-Orford-cedar  
237 Interior ponderosa pine  
247 Jeffrey pine  
256 California mixed subalpine

**SRM (RANGELAND) COVER TYPES :**  
NO-ENTRY

**HABITAT TYPES AND PLANT COMMUNITIES :**

Western white pine is a seral species that is present in a number of habitat types, associations, and communities throughout its range. In northern Idaho and eastern Washington, it may dominate early successional stages of the western hemlock (*Tsuga heterophylla*)/queencup beادلily (*Clintonia uniflora*) and western redcedar (*Tsuga plicata*)/queencup beادلily habitat types [71]. It is also a major seral species in the western hemlock/queencup beادلily habitat type in western Montana and is a major constituent of the western hemlock zone in the Puget Sound area of Washington [11]. A western white pine riparian dominance type has been described for northwestern Montana [14]. Associated species are those associated with the Aralia phase of the subalpine fir (*Abies lasiocarpa*)/queencup beادلily habitat type [14]. Western white pine is moderately abundant, usually growing in small groups and often interspersed with other species, in the subalpine forest zone on the west slope of the Sierra Nevada [50].

## MANAGEMENT CONSIDERATIONS

### SPECIES: *Pinus monticola*

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**WOOD PRODUCTS VALUE :**

Western white pine is highly valued as a timber species. Its wood is straight grained, nonresinous, lightweight, and exhibits dimensional stability. These qualities render the wood useful in the production of window and door sashes. The wood is also used in the production of doors, paneling, dimension stock, matches, and toothpicks [11]. The dimension stock works well. It takes nails without splitting, and it takes a nice finish. The wood is also excellent for carving [1].

**IMPORTANCE TO LIVESTOCK AND WILDLIFE :**

Western white pine provides habitat for a variety of mammals, birds, and insects [3,11,64,66]. Western white pine comprises less than 1 percent of the winter diet of elk [61]; however, it is browsed by black-tailed deer in the winter when other browse is limited [3]. The seeds of western white pine are an important part of the diet of red squirrels and deer mice [11].

**PALATABILITY :**

The palatability of western white pine's foliage to large ungulates is generally rated as poor [3,61,64]. Blue grouse prefer western white pine needles over those of western hemlock [21].

**NUTRITIONAL VALUE :**

The foliar nutrient levels for current year's growth of western white pine were listed as follows [65].

| Percent |      |      |      |      | Parts per million |    |    |     |
|---------|------|------|------|------|-------------------|----|----|-----|
| P       | K    | Ca   | Mg   | S    | B                 | Zn | Fe | Mn  |
| 0.20    | 1.10 | 0.24 | 0.10 | 0.09 | 30                | 45 | 45 | 240 |

**COVER VALUE :**

Western white pine provides nesting, thermal, and foraging cover for a variety of birds [52]; it also provides hiding and thermal cover for elk [19].

**VALUE FOR REHABILITATION OF DISTURBED SITES :**

NO-ENTRY

**OTHER USES AND VALUES :**

Native Americans chewed the resin, wove baskets from the bark, concocted a poultice for dressing wounds from the pitch [62], and collected the cambium in the spring for food [68].

Western white pine forests have aesthetic and recreational value. Cones of western white pine are collected for novelty items [11]. The tree is also planted as an ornamental [26].

**OTHER MANAGEMENT CONSIDERATIONS :**

White pine blister rust: The most serious damaging agent of western white pine is white pine blister rust (*Cronartium ribicola*) [1,11,23]. This rust was introduced into this country at the turn of the century from infected seedlings that had been imported from nurseries in France [1]. White pine blister rust has a life cycle requiring alternate hosts for its completion: five-needled pines and currants (*Ribes* spp.). The rust produces spores on currants that infect white pines. These spores can be dispersed by wind [28] up to 10 miles (17 km) [1]. The spores germinate on the needles, and use the stomatal openings as a vector to the bole of the tree. This usually results in the death of the host tree [28].

Breeding programs have produced strains that are 65 percent resistant to intense exposure to white pine blister rust [2]. Selection of naturally rust-resistant trees for seed sources for natural regeneration and planting rust-resistant nursery stock may keep future damage from blister rust minimal [11]. In established stands that are not rust resistant, thinning tends to increase the number of new lethal infections, while pruning tends to decrease the number of new lethal infections [18].

Fungi: Western white pine is susceptible to three species of needle cast fungi: *Lophodermella arcuata*, *Lophodermium nitens*, and *Bifusella linearis*. It is also susceptible to butt-rot fungi, the most important being *Phellinus pini*, *Phaeolus schweinitzii*, and *Heterobasidion annosum* [11].

The most damaging root disease of western white pine is *Armillaria* spp.,

which causes fading foliage, growth reduction, dead and rotten roots, and black rhizomorphs, resulting in weakened or dead trees [11,70]. Annosus root disease (*Heterobasidion annosum*) also causes some mortality [11]. It spreads radially, infecting an area up to 0.25 acre (0.1 ha) away from stumps [70]. Treating freshly cut stumps with borax has been proven effective in preventing the spread of annosus root disease [22].

**Insects:** Western white pine is susceptible to mountain pine beetle (*Dendroctonus ponderosae*) and emarginate ips (*Ips emarginatus*), and is the principal host for the ips beetle (*Ips montanus*) [11].

**Pole blight:** Pole blight is a physiological disorder brought on by drought. This disease caused significant mortality from 1935 to 1960. Tree mortality was believed to have resulted from rootlet mortality, which reduced western white pine's ability to absorb moisture [11]. The disease is restricted to sites with shallow soils or soils with low moisture retention [30].

**Other:** Western white pine is sensitive to sulfur dioxide and flouride smelter fumes. These contaminants cause the foliage to yellow and drop prematurely. Dwarf mistletoe (*Arceuthobium* spp.) attacks western white pine [11].

**Silvicultural practices:** The method of choice is clearcutting. Selection cutting is not practical because it favors more shade-tolerant species. The composition of a western white pine stand is determined in the first 30 years. Until that time it is fairly plastic, and the stand can be modified by thinning to enhance western white pine growth [11].

**Planting:** Western white pine seedlings are well suited for planting. Both bareroot and container-grown stock exhibit excellent survival and growth [11]. When planting seedlings on droughty sites, it is beneficial to mound the seedbed, as this incorporates organic matter, increases microbial activity, decreases density, and increases the moisture capacity of the soil. This results in increased nutrient availability for seedling growth and increases root penetration [41]. The soil should be packed lightly around the seedling. This practice increases the growth rate in the first year by up to 30 percent [31]. Seedlings planted in fall have a significantly reduced height growth compared with those planted in the spring; however, there is little difference in their survival rates [35].

Nitrogen can be limiting on some sites after harvest. The application of nitrogen at 200 pounds per acre (225 kg/ha) has been found to increase the growth rate of young western white pine stands (less than 10 years old) by 30 percent [37].

**Frost tolerance:** When dormant, western white pine is one of the more frost-tolerant species of the Northwest [36].

**Competitors:** Competing vegetation of western white pine can be effectively controlled by the application of Roundup herbicide (isopropylamine salt of glyphosate). The recommended rate of application is 1 to 3 quarts (1-3 l) of Roundup to 10 gallons (38 l) of aqueous solution per acre (0.4 ha). This treatment had no observable effects on western white pine [33].

## BOTANICAL AND ECOLOGICAL CHARACTERISTICS

**SPECIES: *Pinus monticola***

**GENERAL BOTANICAL CHARACTERISTICS :**

Western white pine is a native, evergreen, long-lived (400+ years), monoecious tree [9,11,16,44]. It can reach 200 feet (60 m) in height and 8 feet (2.4 m) in d.b.h. The needles, 2 to 4 inches (5-10 cm) long, are in bundles of five. The bark on young trees is smooth and grayish green but on mature trees becomes grayish brown, scaly, and separated into rectangular plates [16]. The crown is narrow and composed of regularly spaced branches [1]. In dense stands western white pine self-prunes well, leaving a long, clean bole [16].

The root system consists of a taproot and lateral roots which can spread up to 26 feet (8 m). Most (75 percent) of the lateral roots are in the upper 24 inches (60 cm) of soil [11].

The male strobili are yellow, and the female strobili are reddish purple. Mature female strobili are 5 to 15 inches (12-38 cm) in length [44].

The early growth of western white pine is not rapid, but it is the fastest growing sapling and pole-sized tree in the Northern Rockies [8,13].

**RAUNKIAER LIFE FORM :**

Phanerophyte

**REGENERATION PROCESSES :**

Western white pine can begin producing strobili at 7 years of age [11], but production can be limited by moisture stress and timing. Moisture stress in the early summer of the year strobili mature leads to abortion, while moisture stress in the early summer of the first and second years prior to strobili emergence causes an increase in the number of strobili. Moisture stress in the late summer prior to strobili emergence causes a decrease in strobili numbers [47].

A good crop of female strobili is about 40 per tree [1]. During fair to poor crop years cone beetles (*Conophthorus* spp.), cone moths (*Dioryctria abietivorella* and *Eucosma rescissorianna*), red squirrels, and deer mice can cause partial or complete crop failures [11].

Seed production requires 3 years from the onset of bud initiation. Good seed crops occur every 3 to 4 years. The mean number of seeds per pound is 27,000 (59,000/kg). The seed can be dispersed by wind up to 2,620 feet (800 m) from the parent tree. Seeds remain viable in the duff for up to 4 years, but the germination rate decreases. After 2 years the rate is 25 percent, and after 4 years the rate is 1 percent. Western white pine's seed requires cold moist stratification of 30 to 120 days to germinate; germination is epigeal [11]. Moisture and soil temperature are believed to control the onset of germination. The preferred germination substrate is mineral soil, but seeds will also germinate in duff [8,11].

Seedling mortality is quite high in the first year due to snow mold (*Neoeckia coulteri*), rodents, late season drought, and elevated soil temperatures on dry sites [11]. On dry sites seedling establishment is favored by partial shade, while on moist sites full sunlight favors establishment [11,15].

Vegetative reproduction: Western white pine does not naturally reproduce by sprouting or layering. However, cuttings from young trees treated with rooting hormones (indolebutyric acid) have rooted with fair success [11]. Williams [69] describes the process and considerations for rooting cuttings from older trees (25 years). He obtained a mean success rate of 37 percent.

**SITE CHARACTERISTICS :**

Western white pine is restricted to climates characterized by dry summers and a predominance of winter precipitation [72]. The most extensive and best stands of western white pine are found in the river bottoms and less steep lower slopes of the Priest, Coeur d'Alene, St. Joe, and Clearwater River basins [72]. In British Columbia, western white pine is a minor species on moderately dry to wet, nutrient-medium to nutrient-rich sites in the maritime and subarctic climates [23]. Here, western white pine requires sites fairly rich in calcium and magnesium [25]. However, in the coastal Northwest, western white pine becomes abundant only on poor sites, where it can outcompete Douglas-fir (*Pseudotsuga menziesii*) and other conifers. It does well on unproductive, gravelly soils in the Puget Sound area and reportedly thrives at the edges of bogs on the Olympic Peninsula [1].

**Soils:** Western white pine grows on a wide variety of soils within its range, the majority of which have been classified as Spodosols [11]. Along the West Coast, it attains best development on deep, porous soils, but it is most common on poor, sandy soils. In northern Idaho and other inland sites, it is found on shallow to deep soils, with the surface layers composed of loess or loessial-like material. Parent materials include granite, shist, basalt, and sedimentary rocks. The pH ranges from 4.5 to 6.8 with a mean of 5.4 [11].

**Elevation and topography:** Western white pine is generally a montane species, but grows at a wide range of elevations [11,72]. Elevational ranges vary as follows [11]:

| Area                 | Feet            | Meters        |
|----------------------|-----------------|---------------|
| British Columbia     | 0 to 1,480      | 0 to 450      |
| Vancouver Island, BC | 0 to 3,940      | 0 to 1,200    |
| California           | 6,000 to 10,990 | 1,830 to 3350 |
| Idaho                | 1,540 to 5,910  | 500 to 1,800  |
| Montana              | 1,540 to 5,910  | 500 to 1,800  |
| Oregon               | 6,000 to 7,020  | 1,830 to 2140 |
| Washington           | 0 to 6,070      | 0 to 1,850    |

**Associated species:** In Washington, Oregon, and the Inland Empire, western white pine grows in communities that are rich in other woody and herbaceous flora, but in the Sierra Nevada associated vegetation is usually sparse [11]. In addition to those previously listed under Distribution and Occurrence, overstory associates include Pacific silver fir (*Abies amabilis*), noble fir (*A. concolor*), whitebark pine (*Pinus albicaulis*), foxtail pine (*P. balfouriana*), limber pine (*P. flexilis*), sugar pine (*P. lambertiana*), Jeffrey pine (*P. jeffreyi*), quaking aspen (*Populus tremuloides*), and paper birch (*Betula papyrifera*) [11].

Understory associates include Pacific yew (*Taxus brevifolia*), huckleberry (*Vaccinium* spp.), willow (*Salix* spp.), honeysuckle (*Lonicera* spp.), currant, Rocky Mountain maple (*Acer glabrum*), snowberry (*Symphoricarpos* spp.), ocean-spray (*Holodiscus discolor*), serviceberry (*Amelanchier alnifolia*), pachistima (*Pachistima myrsinites*), sedges (*Carex* spp.), pinegrass (*Calamagrostis rubescens*), false-solomon's-seal (*Smilacina* spp.), wild ginger (*Asarum caudatum*), and queencup beadlily [11].

**SUCCESSIONAL STATUS :**

Western white pine is classified as shade intolerant to very intolerant [24]. It is usually seral to fir (*Abies* spp.), spruce (*Picea* spp.), or hemlock (*Tsuga* spp.) [5,9]. Stickney [59] classified western white pine as a colonizer.

Western white pine does not respond favorably after release from 30 to 60 years of suppression [5,6].

#### SEASONAL DEVELOPMENT :

Height and diameter growth starts from May to late June depending on elevation, aspect, and latitude [11].

Strobili buds emerge in June. The buds are differentiated in July and August of the year preceding emergence. Pollen dispersal lasts for a mean of 8.5 days and usually starts the last week in June. Time of flowering varies over a period of 20 days and is strongly controlled by temperatures during the preceding weeks. It is delayed for 5 days for every 1,000 feet (300 m) gain in elevation, and 6 days per degree Fahrenheit below normal temperatures for May and June. The female strobili ripen from August to September of the second year after bud emergence [11].

The mean phenological development dates for western white pine in northern Idaho were as follows [53]:

| Bark<br>Slips | Shoots<br>Open | Buds<br>Burst | Pollen<br>Starts | Pollen<br>Ends | Shoots<br>End | Winter<br>Buds<br>Formed | Cones<br>Full<br>Size | Cones<br>Open |
|---------------|----------------|---------------|------------------|----------------|---------------|--------------------------|-----------------------|---------------|
| Apr 28        | May 6          | May 21        | Jun 11           | Jun 28         | Aug 11        | Aug 13                   | Aug 1                 | Sep 8         |

## FIRE ECOLOGY

### SPECIES: Pinus monticola

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#### FIRE ECOLOGY OR ADAPTATIONS :

Mature western white pine, with its moderately thick bark (1.5 inches [3 cm]), moderately flammable foliage, height, and evanescent lower limbs, is rated moderate in fire resistance [5,11,56]. However, dense stands, lichen growth, and resinous bark can decrease western white pine's resistance to fire [5].

Young trees with their thin bark are very susceptible to lethal damage by fire [12].

#### FIRE REGIMES :

Find fire regime information for the plant communities in which this species may occur by entering the species name in the [FEIS home page](#) under "Find Fire Regimes".

#### POSTFIRE REGENERATION STRATEGY :

- Tree without adventitious-bud root crown
- Initial-offsite colonizer (off-site, initial community)
- Secondary colonizer - on-site seed
- Secondary colonizer - off-site seed

## FIRE EFFECTS

### SPECIES: Pinus monticola

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**IMMEDIATE FIRE EFFECT ON PLANT :**

Fire of any intensity will damage the cambium layer of young trees, usually resulting in death of the tree [12].

In a mature western white pine stand, a cool fire will kill scattered trees, while only scarring others. However, the fire scars provide a vector for butt rots to enter the tree [46]. Moderate to severe fire in a mature western white pine stand results in cambium damage and crowning, which usually results in the death of the tree [56].

The large amount of humus in western white pine forests renders the trees susceptible to death from heating of the roots [12].

**DISCUSSION AND QUALIFICATION OF FIRE EFFECT :**

NO-ENTRY

**PLANT RESPONSE TO FIRE :**

After a stand-replacing fire, western white pine will seed in from adjacent areas [17]. After a cool to moderate fire that leaves a mosaic of mineral soil and duff, western white pine will reoccupy the site from seed stored in the seed bank [29].

**DISCUSSION AND QUALIFICATION OF PLANT RESPONSE :**

NO-ENTRY

**FIRE MANAGEMENT CONSIDERATIONS :**

Western white pine is a fire-dependent, seral species. Fire exclusion and white pine blister rust have decreased western white pine stocking from 44 percent in 1941 to 5 percent in 1979 [11]. Periodic, stand-replacing fire or other disturbance is needed to remove competing conifers and allow western white pine to develop in early series [5,8,11,67].

Slash burning: Dry sites in the western white pine forest type respond poorly to slash burning, while moist sites respond favorably [24]. Stark [57] provides information on how to estimate nutrient losses from the harvest and slash burning of a western white pine stand. The use of chemical retardant around leave trees in selective cuts has been found to be effective in reducing cambium damage when slash concentrations are light or moderate [51].

For effective fire hazard abatement the recommended Federal slash hazard index is 11 when planning a prescribed fire in western white pine slash. An index of less than 9 will provide little reduction, and greater than 12 generates risk of fire escape [39]. Reinhardt and others [49] provide information on prescribed fire, slash disposal, duff consumption, and management considerations after harvest in western white pine stands in northern Idaho.

Wildlife: Prescribed fire has been recommended in western white pine stands to maintain areas of abundant browse for elk [31].

Other: After wildfire it is recommended that salvage operations begin within the first 2 years [46]. Peterson and Ryan [42] have developed a model based on site, fire, and silvicultural information to predict conifer mortality after wildfire for long-term planning.

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## SPECIES: Pinus monticola

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[FEIS Home Page](#)